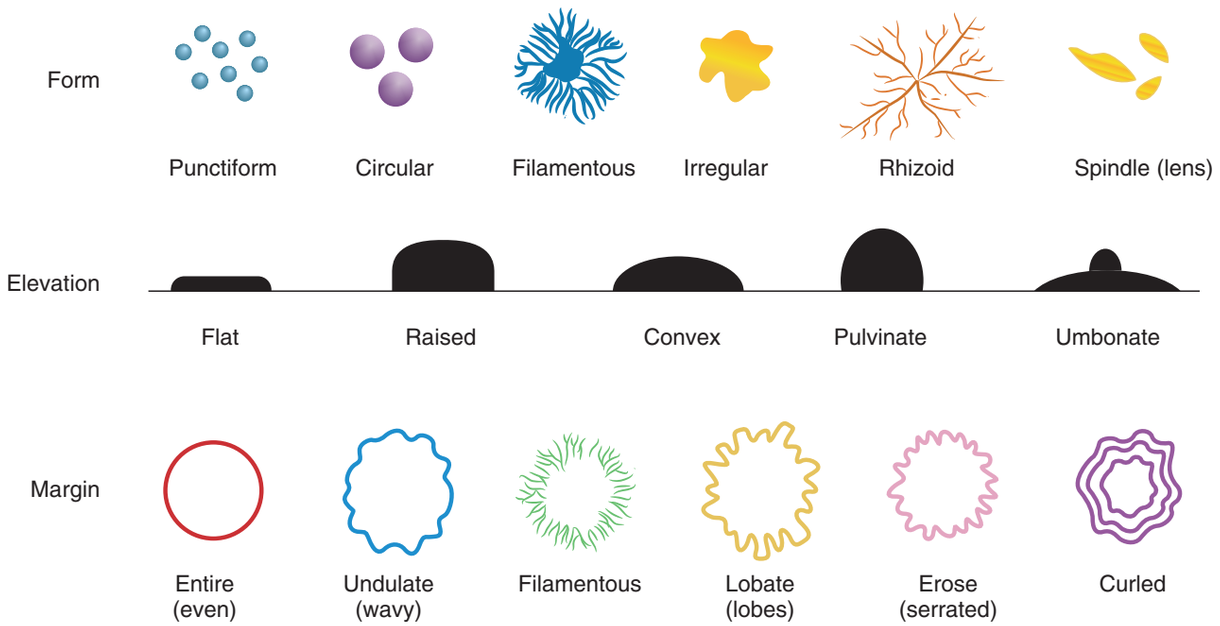


Figure 5.13: Colony morphology of bacteria

5.4.1 Colony Morphology of Bacteria on Solid Media

Shape: The shape of colony may be circular, irregular, filamentous, rhizoid.

Elevation: It is the side view of the colony. It may be flat, raised, umbonate (having a knobby protuberance) crateriform, convex pulvinate (cushion shaped)

Margin: The margin of the bacterial colony may be entire (smooth) irregular, undulate (ovary), lobate, curled, filiform. The irregular shape of the colony give irregular margin (Figure 5.13).

Colony Size: The diameter of the colony is measured in millimeter. It is described in relative terms such as pinpoint, small, medium and large.

Appearance of colony on the surface: The bacterial colonies are frequently shiny/ smooth in appearance. Colonies may be veined, rough, dull, wrinkled, or glistening.

Texture of the colony: Texture means consistency of the bacterial growth. It may be dry, moist, mucoid, brittle (dry breaks apart), viscid (sticks to loop, hard to get off), viscous, or butyrous (buttery).

Opacity of the bacterial Colony: Colonies may exhibit different optical density. It may be transparent (clear), opaque (not clear), translucent (almost clear), or iridescent (changing colour in reflected light).

Colony Odour: Some bacteria produce a characteristic smell, which sometimes helps in identifying the bacteria. Actinomycetes produce an earthy odour which is quite often experienced after rain. Many fungi produce fruity smell while *Escherichia coli* produce a faecal odour.



Smooth colonies of *Streptococcus pneumoniae* are usually virulent, where as rough colonies are non-virulent. But in *Mycobacterium tuberculosis* colonies with rough surface indicates a good factor of virulence.

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